Prediction Depth of Controlled Cellular Automata

A Lean-Verified Classification for One-Sided Rules

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Abstract. We classify the prediction depth of one-sided cellular automata whose frozen control words gate a stay/hop data update over $\mathbb{F}_2$. The input is the explicit initial window and the output is the predicted data bit at its apex. An executable classifier selects a branch for each fixed rule. A single logspace-uniform, polynomial-size De Morgan family for that rule realizes the selected depth bound at every time. Prediction is in logspace-uniform NC^0 exactly when the hop guard is identically zero, or when it is identically one and the stay guard is identically zero. When the hop guard is nonconstant, or is identically one and the stay guard is nonconstant, depth is $\Theta(\log(T+1))$. The homogeneous Pascal rule has depth between the number of set bits of T and three times that number. As a structural comparison, a separate one-path model with transported control has positive-time depth one at equal velocities and exactly $\lceil \log_2(T+1) \rceil$ at unequal velocities. Lean 4 verifies the classification, the connection from dynamics to finite prediction, and the circuit bounds and generators for the frozen rules.

Keywords. Cellular automata, circuit depth, logspace uniformity, Lean 4.

1 Introduction

Algebraic structure can make a cellular automaton's future state accessible to much shallower circuits than layer-by-layer simulation [1]. We ask which fixed guards in a one-sided, two-track automaton admit constant-depth prediction and how depth depends on the observation time.

Our main result is the four-way classification of frozen rules (Theorem 2). A finite guard test selects the branch, and each branch pairs pointwise lower bounds with a single actual generator that meets its upper bound at every time. The homogeneous branch has a nonmonotone depth profile governed by the set bits of the observation time. The main construction turns path counts into parallel prefix statistics and uses those statistics to select coefficients in logarithmic depth (Section 3.4). We also solve a separate one-path rule with transported control as a structural comparison (Theorem 3). There, relative velocity determines how many distinct initial controls the path encounters.

2 Preliminaries

A local rule on configurations over $\mathbb{Z}$ is iterated T times. We ask for one data bit, at site zero. Each model in the main results has a one-sided radius-one update, so its initial window consists of sites $0, \ldots, T$. For the frozen model the input contains $q(T+1)$ control bits followed by $T+1$ data bits. The control width q and the two guard circuits are fixed independently of T. The legal word lengths are $n = (q+1)(T+1)$ and determine T uniquely. The prediction language $L_{s,h}$ consists of the legal words whose predicted bit is one, with all other lengths rejected.

Our upper bounds use Boolean AND and OR gates of fan-in two, NOT gates, and constants. An input wire has depth zero. Gates can share outputs and connect arbitrary sites. The per-time lower bounds apply to any Boolean gates of fan-in at most two in this wiring model. For a fixed rule, write $D(T)$ for the minimum De Morgan circuit depth on the legal window at time T and write $\ell(m) = \lceil \log_2 m \rceil$ for $m \geq 1$. Uniform circuit families are indexed by input length. A deterministic logspace transducer receives the length in unary and writes the complete circuit code, including its gates, wiring, and output. We use logspace uniformity throughout the statements about NC.

A coordinate is *essential* if two assignments that differ only at that coordinate give different outputs.

Essential-input bound. A depth-d circuit of fan-in at most b has at most b^d essential coordinates. For $b \geq 2$, a function with $m \geq 1$ essential coordinates therefore requires depth at least $\lceil \log_b m \rceil$.

Proof. Only input wires in the backwards cone of the output can affect it. At depth zero the cone has at most one input. At the next depth a gate has at most b predecessor cones, so induction bounds the number of input wires by b^d. In a shared circuit, counting a repeated predecessor more than once only increases the bound. □

3 Frozen-control automata

3.1 The iterate formula

Let $a_i \in \{0,1\}^q$ be a control word, $x_i \in \mathbb{F}_2$ an initial data bit, and $s, h : \{0,1\}^q \to \{0,1\}$ the stay and hop guards. The synchronous update is

$$a_i' = a_i, \quad x_i' = s(a_i)x_i + h(a_i)x_{i+1}, \tag{1}$$

where the data arithmetic is in $\mathbb{F}_2$. A backwards path ending at site k has k hops and $T-k$ stays. Put

$$e_k = \prod_{j=0}^{k-1} h(a_j), \quad w_k = \sum_{j=0}^{k} s(a_j), \tag{2}$$

where the second sum is in the natural numbers. For natural w, r, write $M(w,r)$ for the number of weak compositions of r into w parts. For $w > 0$, it is $\binom{w+r-1}{r}$, and $M(0,r)$ is one at $r = 0$ and zero otherwise.

Theorem 1 (Iterate formula). *For every initial configuration and time T, the data bit at site zero is*

$$x_0(T) = \sum_{k=0}^{T} e_k (M(w_k, T-k) \bmod 2) x_k. \tag{3}$$

The same identity holds at any site after translating the indices.

Proof. Expanding the local recurrence once per step expresses the output as a sum over backwards paths. Each hop must be enabled, contributing the factor e_k. A path visits the sites 0 through k in order. The number of stays allocated to each enabled site determines the path, including possible stays after the last hop. There are $M(w_k, T-k)$ such allocations. Their contributions add in $\mathbb{F}_2$, giving Equation 3. The Lean proof establishes the corresponding coefficient recurrence by induction on the iterate of Equation 1. The formula involves only the initial window. □

3.2 Depth classification

Write $p(T)$ for the number of ones in the binary expansion of T, and d_s, d_h for the depths of the fixed guard circuits. A guard varies if it takes both Boolean values. All the bounds below are finite, including $T = 0$.

Guard condition	Lower bound	Uniform upper bound
$h \equiv 0$	0	$1 + d_s$
$h \equiv 1$, $s \equiv 0$	0	0
h varies, or $h \equiv 1$ and s varies	$\ell(T+1)$	$104(\ell(T+3)+1) + \max(d_s, d_h)$
$s \equiv h \equiv 1$	$p(T)$	$3p(T)$

Table 1: Depth bounds at legal lengths for fixed frozen guards.

Theorem 2 (Uniform classification). *For each fixed pair of guard circuits there is one logspace-uniform, polynomial-size De Morgan family that decides the whole language and achieves the appropriate upper bound in Table 1 at every legal length. It has depth $O(\log(n+2))$ at all lengths. The lower column bounds $D(T)$ and holds even for arbitrary fan-in-two Boolean gates. Constant-depth membership is characterized by*

$$L_{s,h} \in \text{logspace-uniform } \mathrm{NC}^0 \quad \Leftrightarrow \quad h \equiv 0 \vee (h \equiv 1 \wedge s \equiv 0). \tag{4}$$

3.3 Proof of the uniform classification

Local and shift cases. For $h \equiv 0$ and $T > 0$, the answer is $s(a_0)x_0$ by idempotence. The pure shift case returns x_T. At $T = 0$ every rule returns x_0.

Varying guards. When h varies, set all data bits except x_T to zero. The surviving term in Equation 3 is $x_T \prod_{j=0}^{T-1} h(a_j)$, regardless of s. A nonconstant Boolean guard has an essential bit, as seen along a single-bit path between two words with different guard values. Each of the first T control words contributes one essential bit on this restriction, and x_T contributes one more. When $h \equiv 1$ and s varies, fix $s(a_0) = 1$ and $s(a_i) = 0$ at sites 1 through T. Then $w_k = 1$ for every k and Equation 3 is parity of all $T + 1$ data bits. Both cases give the same lower bound by essential coordinates.

The upper bound in this branch is given by the construction in Section 3.4.

Homogeneous Pascal rule. For $s \equiv h \equiv 1$, the coefficients are $\binom{T}{k} \bmod 2$. Over $\mathbb{F}_2$, the identity

$$(1+z)^T = \prod_{j:\mathrm{bit}_{j(T)}=1} \left(1 + z^{2^j}\right) \tag{5}$$

shows that precisely the binary submasks of T occur. Their $2^{p(T)}$ data bits are essential. The logarithmic lower bound is therefore $p(T)$. A balanced XOR tree reduces those leaves in $p(T)$ rounds. Each XOR can be implemented with at most three De Morgan layers. The uniform generator scans $k = 0, \ldots, T$, emits the submask wires, and serializes a regular XOR reducer with $5p(T)2^{p(T)}$ gates.

A single uniform family. The guard circuits and the selected branch are fixed. The logspace transducer writes that branch's predictor at legal lengths and a rejecting circuit at other lengths. Its polynomial-length output bounds the number of gates. The local and shift branches have a constant bound. In the varying branch the lower bound grows with T. For the homogeneous branch it grows along the times $T = 2^r - 1$. This proves the constant-depth characterization. □

At $T = 2^r - 1$ the homogeneous rule needs at least r layers, while at the next time $T = 2^r$ it has a circuit of depth at most three. For $r > 3$ this gives a strict drop in minimum depth. The accompanying classifier tests constancy of the guards by traversing control words by prefix. Lean proves that its return value is equivalent to the branch condition. A constancy test stops at its first counterexample and can visit 2^q words in the worst case.

3.4 Uniform prediction circuit

Put $m = T + 1$. After evaluating the two guards at all sites in parallel, the circuit keeps the resulting guard bits and the data track in a register alongside an m-bit one-hot time tag τ. Starting with $u = 0$, the binary digits of T direct a sequence of index doublings and optional increments $u \to 2u$, $u \to u + 1$. Each operation just routes the tag wires, with zero fill. The guard and data tracks pass through. At the end only τ_T is one.

Parallel prefixes. For each k, count the $k + 1$ stay bits to obtain w_k and take the conjunction of the first k hop bits to obtain e_k. All prefixes are computed independently and in parallel. A counter on a bits uses $b = \ell(a + 1)$ bits per word and pads to $P = 2^b$ pairs of words. Each initial pair represents an input bit or zero. A three-word compressor emits the bitwise XOR as its sum word and the bitwise majority shifted left by one as its carry word. These two words preserve the sum modulo 2^b without propagating carries across bit positions. Two such compressors suffice to merge two pairs into one. After b merge

rounds only the root pair contributes. The final addition applies a compressor with third input zero b times, clearing one more low carry bit per step. Since $a < 2^b$, the result is the exact count. The depth is $O(b)$ because there is only one carry-propagating addition on each path.

Coefficient selection. At circuit generation time, enumerate the triples (u, k, j) with $0 \le u, k < m$ and $0 \le j \le k + 1$. Let $C_{u,k,j}$ be zero for $u < k$, and $M(j, u - k) \bmod 2$ otherwise. For $u \ge k$, the $j = 0$ case is one exactly when $u = k$. For $j > 0$, Lucas's theorem tests whether the set bits of $u - k$ are a subset of those of $j + u - k - 1$ [2]. The generator can scan these two integers bit by bit. Each table row with coefficient one tests $w_k = j$ by conjoining matching bit literals, then gates the test with τ_u. A zero row emits a constant. Exactly one candidate pair (u, j) can pass the two tests. Thus the XOR of all the rows for a site is C_{T,k,w_k}, the coefficient in Equation 3. The circuit reduces first over j and then over u, ANDs each result with e_k and x_k, and takes the XOR over k. In this construction each XOR reduction reads the low bit of a carry-save counter.

Stage	Parallel instances	Depth bound
Guard evaluation	$2m$	$\max(d_s, d_h)$
Register initialization and time tag	One register	$6B$
Stay counts and hop conjunctions	m of each	$22B$ and $5B$
Coefficient row tests	$m^2 \frac{m+3}{2}$	$5B$
Parity over j, then over u	m^2, then m	$23B$ each
Gate by hop prefix and data, then final parity	m, then one	$2 + 23B$

Table 2: Resources for the generated frozen predictor, with $B = \ell(T + 3) + 1$. The two prefix computations run in parallel.

Size and generation. The table has $m \sum_{k=0}^{m-1} (k + 2)$ rows, each with $O(\log^2(m + 1))$ gates for the regular conjunction layout. A padded counter has $O(a + 1)$ blocks of $O(\log(a + 2))$ lanes in each of $O(\log(a + 2))$ rounds, with at most $O(\log(a + 2))$ gates per lane even when a compressor is repeated for each output. There are $O(m^2)$ counters including the parity banks, with at most $m + 1$ inputs each. The guard copies, tag routing and conjunctions fit within the resulting $O(m^3 \log^3(m + 1))$ gate bound. The depth is $O(\log(m + 1))$ because only a fixed number of logarithmic-depth stages are composed. Appendix A gives the constants in Table 1.

The serializer counts the unary length n, checks that it is a positive multiple of $q + 1$, and obtains m and T at legal lengths. It enumerates table entries by (k, u, j) and scans their bit positions. The register is indexed in four blocks of width m, for the time, stay, hop and data tracks. Within a counter round, block e uses blocks $2e$ and $2e + 1$ from the preceding round, with zero padding outside the bank. A gate record is determined by the round, block, bit lane and position in its local template. To serialize parallel blocks, the generator counts preceding gates and adds the resulting offset to internal references. In serial composition, it resolves an input reference by regenerating the corresponding predecessor output index. For a chain of width W, the generator appends identity gates $y \wedge y$ so the outputs of each stage have consecutive indices. With N primary inputs and G preceding gates, a local wire number a then becomes $N + G - W + a$. This lets the generator count through the rounds without a stack. The nesting of the bank serializers has fixed depth. The loop indices, offsets and references are all polynomially bounded in n, so a fixed number of counters and arithmetic temporaries suffice using $O(\log(n + 2))$ bits of workspace. The gate and output lists are streamed.

4 Transported controls as a structural comparison

In this separate model the control track has its own dynamics and the data follows one path. Fix velocities $u, v \in \{0, 1\}$ and a Boolean control and data track (c_i, d_i). Define the synchronous rule

$$c'_i = c_{i+u}, \quad d'_i = c_i \wedge d_{i+v}. \tag{6}$$

The window has $2(T+1)$ bits. With a single backwards data path, the data at site i and time T equals

$$d_{i+vT} \wedge \bigwedge_{k=0}^{T-1} c_{i+kv+(T-1-k)u}, \tag{7}$$

where the right-hand side uses initial bits and the conjunction is empty at $T = 0$.

Theorem 3 (Exact depth under transport). *Depth is zero at $T = 0$. For $T > 0$, it is exactly one if $u = v$ and exactly $\ell(T+1)$ if $u \neq v$. The formal upper-bound construction uses a padded AND tree with at most $2m$ binary AND gates for m essential inputs. The lower bound applies to all fan-in-two Boolean circuits predicting the data bit.*

Proof. At the k-th step of the backwards path we read control at site $i + kv$ and time $T - 1 - k$. The initial site of that control bit is $i + kv + (T - 1 - k)u$, which proves Equation 7. At $T = 0$ the answer is the input wire d_i. At positive times with equal velocities all control indices coincide, leaving a conjunction of one control bit and one data bit. In general, the initial control index is $i + (T-1)u + k(v-u)$. For unequal velocities the indices are $i, \ldots, i + T - 1$ in forward or reverse order. Every distinct factor is essential, since flipping it with all other factors set to one changes the output. Thus there are two essential coordinates in the equal case and $T + 1$ in the unequal case. For m factors, a balanced AND tree of height $\ell(m)$ attains the essential-input lower bound. Padding to the next power of two uses $2^{\ell(m)} - 1 \leq 2m$ binary gates. □

The pair $(u, v) = (0, 0)$ is the frozen no-hop case with stay guard c_i. Changing only u to one gives logarithmic depth although the data never hops. For $(u, v) = (1, 1)$ the data and control both move, yet their equal velocity gives constant depth. For this single-path rule, the relevant count is the number of distinct initial controls encountered in the moving frame of the data path. The time-dependent framework in [1] accommodates this rule. Its single path makes the exact depth accessible.

5 Related work

Pascal parity and fast prediction. The odd-entry count $2^{p(T)}$ is classical. Rowland attributes the binary count to Glaisher and derives it from Lucas's theorem [2]. The fast cellular-automaton prediction work in [1] already uses the Pascal coefficients for homogeneous additive rules. Its modular-sum gates can have unbounded fan-in. The Pascal claim here is a per-time bounded-fan-in lower bound and a matching profile up to a constant factor with a verified uniform De Morgan generator.

Dependent linear tracks and the coefficient formula. Moore's general divide-and-conquer method for time- and space-dependent linear stages gives an $O(\log^2(T+2))$ bounded-fan-in prediction bound in the stated setting [1]. Freezing the controls and using the one-sided stay/hop stencil lets us use prefix statistics and reach $O(\log(T+2))$ in this specialization. The coefficient in Equation 3 also has a classical matrix analogue. The one-step data map is upper bidiagonal on the relevant finite window. Higham's divided-difference formula for functions of such matrices gives the product of intervening off-diagonal entries times a divided difference in the diagonal entries [3]. For the monomial z^T this divided difference is a complete homogeneous polynomial of degree $T - k$. On zero-one diagonal entries it counts the same weak compositions. The polynomial identity can be specialized to integer zero-one entries before reducing modulo two, with no division in $\mathbb{F}_2$.

Mechanization and the meaning of frozen. Silváši and Tomášek formalized bounded grids and computable cellular automata in Lean [4]. Their work supplies precedent for machine-checked cellular automata. The present artifact checks a rule-family depth classification and the actual circuit generator. The freezing model of Ollinger and Theyssier imposes a monotone order on each whole cell state [5]. In Equation 1 only the control word is fixed, and the data at a site can cycle. The contribution here is the combination of an executable rule classifier, pointwise depth bounds and a verified uniform generator for the full frozen rule family. The exact four-way classification and the one-path transport specialization have undetermined priority in the literature.

6 Mechanization and scope

The window interface separates infinite dynamics from finite inputs. `rawSample` reads the window from a configuration, and `decodeRaw` extends an arbitrary legal window to an infinite configuration. Lean proves that sampling this extension recovers the supplied bits. Correctness on sampled configurations therefore gives correctness on every legal input. Lifting from time to input length selects that same predictor at legal lengths and a rejecting circuit at the others.

The uniformity proof generates the code of the circuit used for the depth bound, including the variable-length lists of gates and output references. The reusable code combinators verify reference bounds and relocation for parallel banks and serial composition. These obligations allow a generator to recover individual references by recomputation in logarithmic space. The classification theorem uses the selected circuit family as one witness for correctness, size, uniformity and the per-time bound. The transport theorem establishes optimal depth separately at each window length.

Each frozen control word selects one of four local data maps 0, x_i, x_{i+1} and $x_i + x_{i+1}$. Once the controls are fixed, the data dynamics is one-sided and linear. The input is an explicit window with length linear in T, and the circuit can wire across arbitrary sites. The accompanying development also formalizes composition and simulation criteria for other models. Their application requires a separate implementation or reduction witness.

7 Conclusion

The exact NC^0 criterion and the Pascal profile describe distinct ways that frozen prediction depth depends on time. For varying guards, prefix counts make the path coefficients accessible in logarithmic depth, which lets the same uniform family attain the upper bound at each legal length. The transported rule exhibits another mechanism. Relative motion changes depth by changing how many initial control bits a single data path reads.

Artifact availability

The formal results refer to this fixed Lean source revision. The Typst source and build instructions are in the project repository. Use the full commit SHA associated with the PDF to select the source snapshot. The repository documents the pinned Typst and font inputs for rebuilding all four editions. GitHub Actions verifies that all four PDFs rebuild from the checked-out commit. Run `./scripts/check.sh` for the strict Lean build and axiom audit. Appendix B lists the declarations behind the main results.

A Depth budget

The normalized counter tree costs at most $2 + 14b$ layers, including initialization and copying in each merge round. The finishing addition costs at most $1 + 8b$. Together they cost at most $22(b + 1)$. The counts therefore cost at most $22B$, and taking the low bit of a counter for parity is bounded by $23B$ including the empty case. The depth proof charges $5B$ for a row test, from the literal comparisons, conjunction and time tag. The hop conjunction, bounded by $5B$, runs alongside the lookup. The decoder budget is $22B + 5B + 3 \cdot 23B + 2 = 96B + 2 \leq 98B$. Register initialization costs at most two layers with normalization. There are at most $2\ell(T + 1)$ subsequent register stages of at most two layers each, also counting normalization. This gives at most $6B$ for the register path and a total of $104B + \max(d_s, d_h)$ including guards.

B Lean correspondence

The paths below are relative to `PredictionDepth/`.

Result	Lean module and declaration
Frozen iterate	`FrozenPascal/Semantics.lean` `iterate_data`
Window interface	`FrozenPascal/GuardRules.lean` `rawSample_decodeRaw`
Executable classifier	`FrozenPascal/Classification.lean` `classify`, `classify_eq_iff`

Uniform classification	`FrozenPascal/UniformClassification.lean`
	`uniform_depth_classification`
	`predictionLanguage_mem_uniformNC_zero_iff`

Uniform predictor	`FrozenPascal/UniformPrediction.lean`
	`uniformGuardedPrediction_code_logspace`
	`uniformGuardedPrediction_depth`

Code composition	`Complexity/CodeLogspace.lean` `compCode_logspace`
Pascal support	`FrozenPascal/HomogeneousPascal.lean` `pascalSupport_card`

Transported control	`TransportedControl.lean`
	`iterate_data_true_iff`, `essential_prediction_iff`
	`exact_prediction_depth`, `constant_depth_iff`

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